

NEBULA Reconstruction Quality Study

Part A — Source-Level Traceback of the Double Peak in Δp_x

Part B — (p_x, p_y) Efficiency Scan

Baiting Tian

May 13, 2026

Contents

1	Introduction	1
2	Part A: Source-Level Traceback of the Double Peak in Δp_x	2
2.1	Definition of the Quantity	2
2.2	Reproduction: Lab Frame vs. Rotated Frame	2
2.3	Layer-by-Layer Data Flow	3
2.4	Key Split Plot	3
2.5	Discrete Bands in NEBULA Reconstructed x	4
2.6	Conclusion and Future Options	4
3	Part B: Truth-Only (p_x, p_y) Efficiency Scan	5
3.1	Scan Configuration	5
3.2	Geometry Acceptance Method	6
3.3	Two-Dimensional Efficiency Maps	7
3.4	$p_y = 0$ Slice	8
3.5	Conditional Efficiencies: Bottleneck Identification	8
3.6	$\varepsilon_{\text{det}} > \varepsilon_{\text{geom}}$ Is Not a Bug: Two Edges Deviate from $\varepsilon_{\text{det}} = \varepsilon_{\text{geom}} p_{\text{direct}}$	9
3.7	2D Decomposition of $\varepsilon_{\text{direct}}$ and $\varepsilon_{\text{cascade}}$	12
3.8	Forward Acceptance Bias on R (Not an Unfolding)	13
3.9	Part B Conclusions	15
4	Conclusions	15
4.1	Part A Summary	15
4.2	Part B Summary	15
4.3	Future Work	16

1 Introduction

`nn.breakup_phys.20260503.zh.tex` §5.4 reported a clear double peak (central dip) in the NEBULA neutron reconstruction residual Δp_x . The first hypothesis was that the structure came from the 12 cm bar pitch of NEBULA along X combined with the downstream `n_reco_neutrons==1` filter. The same report also observed that the NEBULA hit rate in the tight-cut subset was only 18–22% for ypol, but it did not quantify how this hit rate varies across (p_x, p_y) .

This report closes both open points as an independent investigation. Part A uses the existing 4.59M-event ypol breakup CSV to trace the double-peak structure back through the reconstruction source code and provides the key split plots. Part B runs a truth-only neutron-gun scan,

decomposes the NEBULA acceptance into geometry, detection, and reconstruction layers, and quantifies them as functions of (p_x, p_y) . This study is **read-only**: it does not modify either NEBULARECO or the converter code.

2 Part A: Source-Level Traceback of the Double Peak in Δp_x

2.1 Definition of the Quantity

The “double peak in Δp_x ” in §5.4 is a **per-particle reconstruction residual**:

$$\Delta p_x \equiv p_x^{\text{truth},n} - p_x^{\text{reco},n},$$

the difference between truth and reconstructed momentum for the same neutron. This is **not** the event-level R observable used in the ypol polarization analysis,

$$\Delta p_x^{\text{rot}} = p_x^{\text{p,rot}} - p_x^{\text{n,rot}}.$$

The former is NEBULA-neutron-only and probes reconstruction error; the latter combines the proton and neutron and probes the physics asymmetry.

Data source: ypol 16-cell breakup g4output from spana03, processed with NN proton reconstruction plus NEBULA neutron reconstruction. `02_extract_observables.C` extracts the observables, and `03_analyze_r_breakup.py` adds the reaction-plane rotation to produce `joined.parquet` (3.38M events, 1.20M-event NEBULA-hit subset).

2.2 Reproduction: Lab Frame vs. Rotated Frame

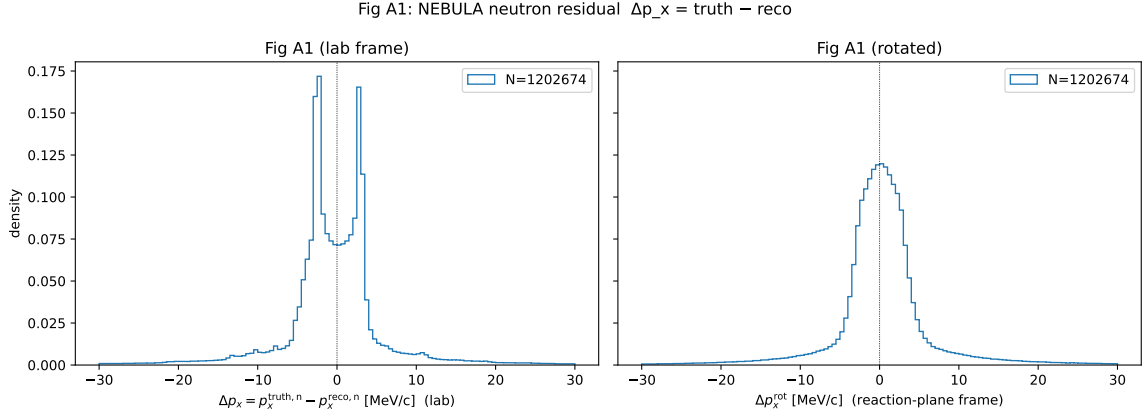


Figure 1: Fig A1: Neutron residual $\Delta p_x = p_x^{\text{truth},n} - p_x^{\text{reco},n}$ in the NEBULA-hit subset (1.20M events). **Left (lab frame)**: a clear double peak at $\pm 2\text{--}3$ MeV/c, with a central dip near zero, consistent with §5.4. **Right (after reaction-plane rotation)**: a single peak, with no double-peak structure. The reason is that the reaction-plane rotation moves each event’s lab-frame discrete peaks to different ϕ angles, so the discreteness is averaged away event by event.

Main point: the double peak described in §5.4 is visible only in the lab-frame Δp_x . The event-level R observable, which uses the rotated frame proton-neutron difference, **does not show** this double peak because (a) the rotation smears out the neutron’s lab-frame discreteness and (b) the NN proton contributes an additional continuous broadening. Therefore the concern that the double peak directly biases the R measurement is, at least in the direct sense, not supported.

2.3 Layer-by-Layer Data Flow

L1 — Geant4 SimData layer. When the Geant4 sensitive detector deposits energy in a NEBULA Neut bar, it writes the continuous `fPrePosition` into `TSimData`. This is the `simdata`-level x distribution and is not discrete. Source: `libs/smg4lib/src/data/src/NEBULASimDataConverter_TArtNEBULA`

L2 — The converter replaces X by the bar geometry center. In `NEBULASimDataConverter_TArtNEBULA` the code builds `TArtNEBULAPla` with

```
Double_t x = prm->fPosition.x() + NEBULAParameter->fPosition.x();
```

`prm->fPosition.x()` is the geometric center of the bar in the NEBULA local coordinate system. Along X the pitch is ≈ 122 mm, with 60 bars per layer $\times 2$ layers = 120 Neut bars covering $x \in [-1962, +1962]$ mm. Therefore the hit-level x distribution consists of 60 discrete peaks with a 12 cm pitch, while y remains continuous because it is computed from the two-ended PMT time difference, $0.5 \text{ dt } v_{\text{scinti}}$, at line 159.

L3 — The 5 mm Gaussian smearing in the reconstructor is far too small. `NEBULAReco.cc:295-305` applies `ApplyPositionSmearing`, adding a Gaussian jitter with $\sigma = 5$ mm in x , y , and z . Since 5 mm $\ll$ the 60 mm half-width of a bar, this barely removes the discreteness. The expected scale is

$$\Delta p_x = -\frac{\Delta x p_z}{L} \in \pm \frac{60 \text{ mm} \cdot 600}{7250} \approx \pm 5 \text{ MeV}/c,$$

using $L_{\text{NEBULA}} \approx 7.25$ m and a typical neutron lab-frame p_z . Fig A1 shows peaks around $\pm 2-3$ MeV/ c , consistent with the RMS scale rather than the full half-bar edge.

L4 — Cluster energy-weighted centroid. In `NEBULAReco.cc:206-218`, the cluster position is computed as

```
weightedPos += weight * hit.position;
totalWeight += weight;
```

For a cluster spanning multiple bars, the centroid is the energy-weighted average of the participating bar centers. However, the participating bars are still drawn from a discrete set of centers, so the centroid solution space is still finite and discrete rather than continuous. Thus “multi-bar clusters de-discretize the position” is only partially true: the centroid softens the single-bar quantization but does not make the result continuous.

L5 — The key split by `hit_mult_n`. `n_reco_neutrons` counts the number of clusters, not the number of bars. A multi-bar cluster is still a single `RecoNeutron`. To distinguish these cases, this Part A task added `hit_mult_n = re->neutrons[0].hitMultiplicity` to `02_extract_observables.C`; it is the number of bars in the first `RecoNeutron` cluster. In the NEBULA-hit subset, single-bar clusters (`mult=1`) are 43%, while multi-bar clusters (`mult  $\geq 2$`) are 57%.

2.4 Key Split Plot

Important negative result: the §5.4 hypothesis that the central dip is caused by the `n_reco_neutrons==1` filter removing multi-bar events is **incorrect**. The split gives two independent checks:

1. `n_reco_neutrons==1` does not distinguish the number of bars inside a cluster. A multi-bar cluster is still one `RecoNeutron` and passes the filter.
2. Fig A2 shows that both the single-bar and multi-bar subsets have the same double-peak shape; the multi-bar subset does not fill the center.

The correct conclusion is that the double peak is an intrinsic discrete structure of the **lab-frame neutron residual** and is only weakly related to cluster topology. The “central dip” seen in §5.4 is simply the density shape of a double peak produced by a uniform position inside a single bar; it is not introduced by the filter.

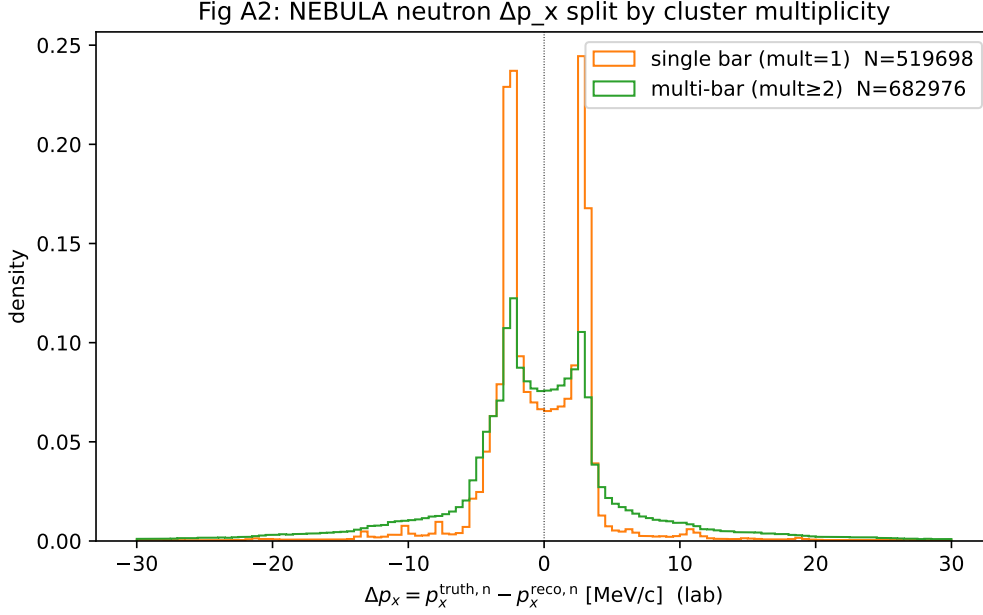


Figure 2: Fig A2: Lab-frame Δp_x in the NEBULA-hit subset (1.20M events), split by `hit_mult_n`. Single-bar clusters (`mult=1`, 520k events) and multi-bar clusters (`mult≥2`, 683k events) **both** show a double-peak structure, with the same outer-peak positions at $\pm 2\text{--}3$ MeV/ c . The multi-bar case has lower outer peaks and broader tails, so the weighted centroid does soften the boundary behavior slightly, but it **does not** flatten the double peak.

2.5 Discrete Bands in NEBULA Reconstructed x

2.6 Conclusion and Future Options

Conclusion: the lab-frame double peak in Δp_x comes from the 12 cm NEBULA bar discretization along x plus an insufficient 5 mm smearing. The §5.4 hypothesis that “the filter removes multi-bar events and creates the central dip” is rejected by two independent observations: (a) `n_reco_neutrons==1` does not distinguish the bar multiplicity inside a cluster; (b) the single-bar and multi-bar subsets have the same double-peak shape. The double peak is a discrete structure of the lab-frame residual itself and is not caused by cluster topology. Once transformed into the reaction-plane frame, the double peak is averaged away by the event-by-event rotation, and the event-level R observable does not show it.

Current choice: leave the reconstruction code unchanged to preserve continuity with the elastic report and the existing `ypol R` values.

Three future options if the discreteness must be removed:

1. **Increase X smearing to ≥ 60 mm:** set a separate $\sigma_x \approx 60$ mm in `ApplyPositionSmearing`.
Cost: this artificially removes the bar-position information and is less faithful to the real detector response.
2. **Keep multi-bar clusters and change the downstream filter:** the current `n_reco_neutrons==1` already includes multi-bar clusters, so it is not the problem. If high-multiplicity events are a concern, explicitly apply an upper bound to `hit_mult_n` (for example ≤ 5) instead of using the cluster count. This is a small analysis-side change.
3. **Sub-bar reconstruction:** interpolate x inside a bar using the energy-deposit pattern.
Cost: this requires `simdata`-level information and a larger reconstruction-chain change, and it also deviates from the ANAROOT quantity convention.

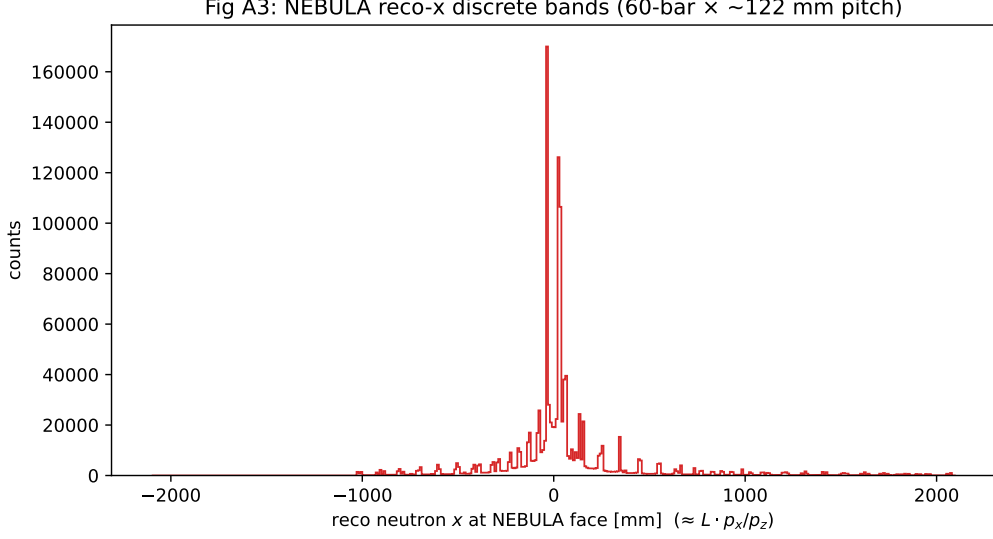


Figure 3: Fig A3: Reconstructed neutron x estimate at the NEBULA plane for the NEBULA-hit subset, $x \approx L p_x^{\text{reco}} / p_z^{\text{reco}}$ with $L = 7250$ mm. About 60 equally spaced peaks appear, with pitch ≈ 122 mm, matching the bar centers in `NEBULA_Detectors_Dayone.csv`. Single-bar clusters sit exactly on the peaks; multi-bar clusters move the centroid and partially fill the valleys.

3 Part B: Truth-Only (p_x, p_y) Efficiency Scan

3.1 Scan Configuration

Execution host: labenpg (96 cores, ENPG host case). `sim.deuteron` ran 32 shards in parallel, followed by NEBULA-only reconstruction (`run_nebula_reco.C`).

Gun: single neutron, emitted from the `TBeamSimData` tree. The position is the target position in the mac file, $(-12.488, 0.009, -1069.458)$ mm in lab coordinates. The momentum grid is defined in the target frame (target rotated by $+3^\circ$ about Y): $p_x^{\text{tgt}} \in [-200, +200]$ MeV/c in 20 MeV/c steps, $p_y^{\text{tgt}} \in [-100, +100]$ MeV/c in 10 MeV/c steps, and $p_z^{\text{tgt}} \in \{550, 600, 627, 650, 700\}$ MeV/c. The five p_z points are equally marginalized. Each subpoint has 2000 events, for a total of $21 \times 21 \times 5 \times 2000 = 4,410,000$ events, split into 32 shards with about 137813 events per shard. For each event the target-frame momentum is rotated to the lab frame by $R_y(+3^\circ)$ and written to `TBeamSimData.fMomentum`; `sim.deuteron` receives the lab-frame momentum.

Geometry: reuse `configs/simulation/geometry/3deg_1.15T.mac` (30° dipole, 1.15 T field map `180703-1,15T-3000.table`) plus `Target/SetTarget false`. The target material is disabled to avoid target-scattering contamination in the acceptance measurement, while the target position remains the mac-file source of truth and the gun starts from that position.

NEBULA configuration: Day-one NEBULA only; NEBULA Plus is not enabled. `3deg_1.15T.mac` loads only `NEBULA_Dayone.csv` and `NEBULA_Detectors_Dayone.csv`. This is the SAMURAI Day-one NEBULA array: 2 walls $\times$ 2 sub-layers $\times$ 30 bars = 120 Neut bars plus 24 Veto bars, with global position $(0, 0, 7249.72)$ mm. This repository contains no NEBULA Plus parameter files or code paths. Therefore all $\varepsilon_{\text{geom}}$, ε_{det} , and $\varepsilon_{\text{reco}}$ scan results in this report are Day-one-only results and do not include the forward-acceptance gain from NEBULA Plus modules.

Three efficiency definitions. For each target-frame grid point $(p_x^{\text{tgt}}, p_y^{\text{tgt}})$, the five p_z subpoints are pooled (5 p_z values $\times$ 2000 events = 10000 events per bin). The numerators and denominators are:

Table 1: Numerator and denominator definitions for $\varepsilon_{\text{geom}}$, ε_{det} , and $\varepsilon_{\text{reco}}$. $N_{\text{total}} = 10000$ is the number of truth-neutron-gun events in each (p_x, p_y) bin.

Efficiency	Numerator (event count)	Denom
$\varepsilon_{\text{geom}}$	$N_{\text{geom}} = \text{ray crosses any NEBULA Neut bar AABB}$ (pure Python ray-cast, <code>geom_acceptance.py</code> ; see §3.2)	N_{total}
ε_{det}	$N_{\text{det}} = \text{NEBULAPla total energy } \sum_k Q_k^{\text{AveCal}} > 1 \text{ MeV}$ (Geant4 neutron-plastic reaction + 1 MeV threshold; <code>run_nebula_reco.C</code> output <code>nebula_sumE</code>)	N_{total}
$\varepsilon_{\text{reco}}$	$N_{\text{reco}} = \text{n_reco_neutrons} \geq 1$ (at least one <code>RecoNeutron</code> after NEBULAReco time clustering)	N_{total}

Conditional efficiencies, after cancelling the common denominator, are

$$\begin{aligned}\varepsilon_{\text{det}}|\text{geom} &= \varepsilon_{\text{det}}/\varepsilon_{\text{geom}} = N_{\text{det}}/N_{\text{geom}}, \\ \varepsilon_{\text{reco}}|\text{det} &= \varepsilon_{\text{reco}}/\varepsilon_{\text{det}} = N_{\text{reco}}/N_{\text{det}}.\end{aligned}$$

$\varepsilon_{\text{det}}|\text{geom}$ is the probability that a neutron entering the active volume actually produces enough energy deposition (neutron-plastic reaction probability times threshold passing rate). $\varepsilon_{\text{reco}}|\text{det}$ is the probability that a detected NEBULA hit pattern is successfully clustered. The implementation is in `scripts/analysis/nebula_reco_quality/analyze_efficiency.py`.

3.2 Geometry Acceptance Method

$\varepsilon_{\text{geom}}$ is computed with a **pure Python ray-AABB intersection test**, not with Geant4. Reasons: (i) this separates pure geometry acceptance from detection and reconstruction effects; neutron-bar reactions, Q thresholds, and clustering do not enter this layer; (ii) the geometry CSV files are the same configuration read by the Geant4 simulation (`NEBULA_Detectors_Dayone.csv` plus `NEBULA_Dayone.csv`), so $\varepsilon_{\text{geom}}$ and the simulation geometry are strictly sourced from the same files; (iii) the 4.41M event-level geometry check finishes in about 10 seconds in pure Python, about 10^3 faster than Geant4 ray tracing.

Bar geometry The NEBULA Neut section contains 120 plastic scintillator bars, with geometry from `NEBULA_Detectors_Dayone.csv`:

- Each bar has a $120 \times 120 \text{ mm}^2$ cross section and a 1800 mm long axis along Y (the `NeutSize` row).
- Four sub-layers are arranged at local $z \in \{0, 130, 846, 976\}$ mm. Each sub-layer contains 30 bars along X with a center pitch of ≈ 122 mm. Layer 1 = sub-layers 1+2 = front 60 bars; Layer 2 = sub-layers 3+4 = rear 60 bars.
- The global NEBULA `Position` is (0, 0, 7249.72) mm (first row of `NEBULA_Dayone.csv`).

Each bar is represented in the lab frame as an AABB:

$$\text{bar}_i = [c_x^{(i)} \pm 60, c_y^{(i)} \pm 900, c_z^{(i)} \pm 60] \text{ mm},$$

where (c_x, c_y, c_z) is the local bar position from the detectors CSV plus the global `Position`.

Ray model For a neutron event with target-frame momentum $(p_x^{\text{tgt}}, p_y^{\text{tgt}}, p_z^{\text{tgt}})$, the momentum is first rotated by $R_y(+3^\circ)$ to obtain $(p_x^{\text{lab}}, p_y^{\text{lab}}, p_z^{\text{lab}})$, matching the gun input used by `sim.deuteron`. The lab-frame ray is

$$\vec{r}(t) = \vec{O} + t(p_x^{\text{lab}}, p_y^{\text{lab}}, p_z^{\text{lab}}), \quad t \geq 0, \quad \vec{O} = (-12.488, 0.009, -1069.458) \text{ mm}.$$

The origin $\vec{O}$ is the target position in the mac file, matching the sim-gun position. If $p_z^{\text{lab}} \leq 0$, the event is immediately rejected.

Intersection test (slab method) The standard AABB ray-intersection test is used. For each axis $a \in \{x, y, z\}$, compute the ray parameters for entering and leaving the bar's slab, t_a^{in} and t_a^{out} . If $|p_a^{\text{lab}}| < 10^{-12}$ and O_a is outside the bar on that axis, reject the bar immediately. Then take $t_{\text{near}} = \max_a t_a^{\text{in}}$ and $t_{\text{far}} = \min_a t_a^{\text{out}}$. If $t_{\text{near}} \leq t_{\text{far}}$ and $t_{\text{far}} \geq 0$, the ray hits that bar. An event is marked as hit if *any* bar is hit:

```
ox, oy, oz = -12.488, 0.009, -1069.458 # mac target position [mm]
# px, py, pz are lab-frame momentum (target frame rotated by R_y(+3deg))
for bar in bars: # 120 NEBULA Neut bars
    if ray_hits_bar(ox, oy, oz, px, py, pz, bar):
        hit = 1
        break
```

Grid and p_z marginalization The scan evaluates one hit flag (0/1) on each of the $21 \times 21 \times 5 = 2205$ grid points, producing `eps_geom_grid.parquet` with columns $(p_x, p_y, p_z, \text{hit})$. `analyze_efficiency.py` averages over the five p_z subpoints for each (p_x, p_y) :

$$\varepsilon_{\text{geom}}(p_x, p_y) = \frac{1}{5} \sum_{p_z \in \{550, 600, 627, 650, 700\}} \text{hit}(p_x, p_y, p_z).$$

Known approximations and edge effects

1. The gun position and ray origin both use the mac-file target position $(-12.488, 0.009, -1069.458)$ mm, matching the real experimental target position. The target material is disabled with `Target/SetTarget false` to prevent target scattering from contaminating the acceptance measurement. **An earlier version used $\vec{O} = (0, 0, 0)$; this was corrected in commit a0d769c.** The present result uses the physically correct target position.
2. Momentum input is target-frame momentum, rotated to the lab frame by $R_y(+3^\circ)$ for Geant4 and ray casting. The target-frame and lab-frame treatments are therefore consistent. In the downstream §3.6 *R* check, the lab-frame truth from `joined.parquet` is rotated back to the target frame with $R_y(-3^\circ)$.
3. The AABB model treats each bar as an ideal rectangular box. Real sub-layers have mm-scale air gaps (local $z \in \{0, 130, 846, 976\}$ mm). The effect on $\varepsilon_{\text{geom}}$ is below 1% and is ignored.
4. Neutron absorption or scattering inside a bar is not included in $\varepsilon_{\text{geom}}$. That part is measured separately by ε_{det} using Geant4 plus the 1 MeV threshold. $\varepsilon_{\text{geom}}$ only asks whether the ray enters the active volume.

3.3 Two-Dimensional Efficiency Maps

The panel definitions in Fig B1 are:

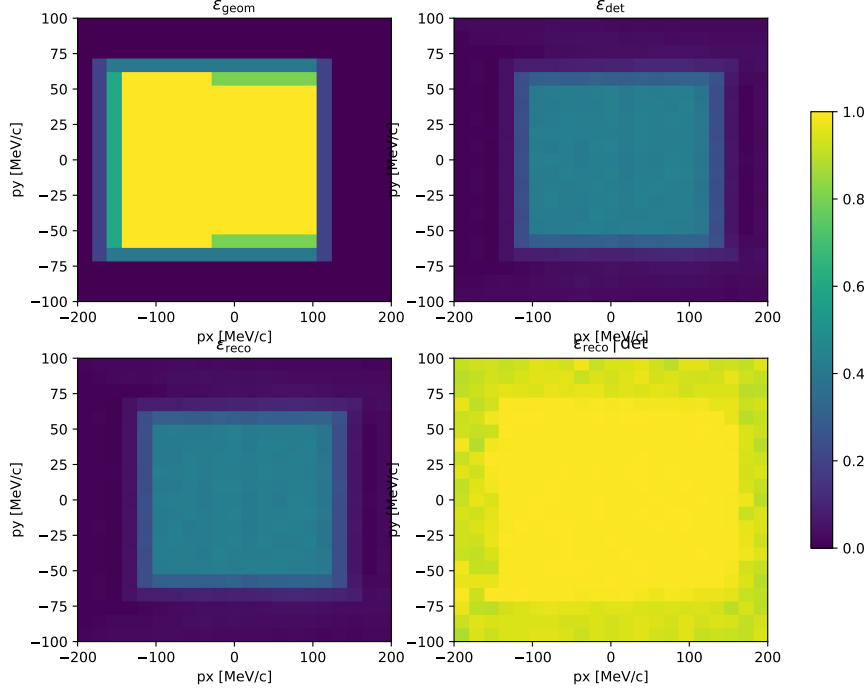


Figure 4: Fig B1: Four efficiency maps in target-frame $(p_x^{\text{tgt}}, p_y^{\text{tgt}})$, with a shared $[0, 1]$ colorbar. Panel numerators and denominators are defined below. $\varepsilon_{\text{geom}}$ is an asymmetric rectangle: $p_x^{\text{tgt}} \in [-180, +120]$ MeV/c (target 3° rotation plus target position $(-12.488, 0.009, -1069.458)$ mm shifts the NEBULA center toward $p_x^{\text{tgt}} < 0$), and vertical extent $|p_y^{\text{tgt}}| \leq 65$ MeV/c. ε_{det} and $\varepsilon_{\text{reco}}$ have the same shape but lower absolute scale ($\sim 20\%$), equal to the neutron-plastic detection efficiency ($\sim 43\%$) times $\varepsilon_{\text{geom}}$. $\varepsilon_{\text{reco}}|\text{det}$ is nearly 1.0 everywhere.

- $\varepsilon_{\text{geom}} = N_{\text{geom}}/N_{\text{total}}$: denominator = incident truth-neutron events ($N_{\text{total}} = 10000/\text{bin}$); numerator = events whose ray crosses any NEBULA Neut bar AABB.
- $\varepsilon_{\text{det}} = N_{\text{det}}/N_{\text{total}}$: same denominator; numerator = events with Geant4-simulated NEB-ULAPla $\sum Q > 1$ MeV.
- $\varepsilon_{\text{reco}} = N_{\text{reco}}/N_{\text{total}}$: same denominator; numerator = events with NEBULAReco `n_reco_neutrons` ≥ 1 .
- $\varepsilon_{\text{reco}}|\text{det} = N_{\text{reco}}/N_{\text{det}}$: denominator = detected subset N_{det} ; numerator = the detected events that also pass reconstruction.

3.4 $p_y = 0$ Slice

3.5 Conditional Efficiencies: Bottleneck Identification

The two Fig B3 curves use **different denominators** from Fig B2:

- $\varepsilon_{\text{det}}|\text{geom} = N_{\text{det}}/N_{\text{geom}}$ (blue): denominator = geometrically accepted events N_{geom} whose ray enters the active volume; numerator = events with $\sum Q > 1$ MeV. This measures the neutron-plastic interaction plus the 1 MeV threshold passing rate, decoupled from geometry.
- $\varepsilon_{\text{reco}}|\text{det} = N_{\text{reco}}/N_{\text{det}}$ (orange): denominator = detected events N_{det} ; numerator = those that also pass reconstruction. This measures the loss of the NEBULA cluster reconstruction chain on already detected events.

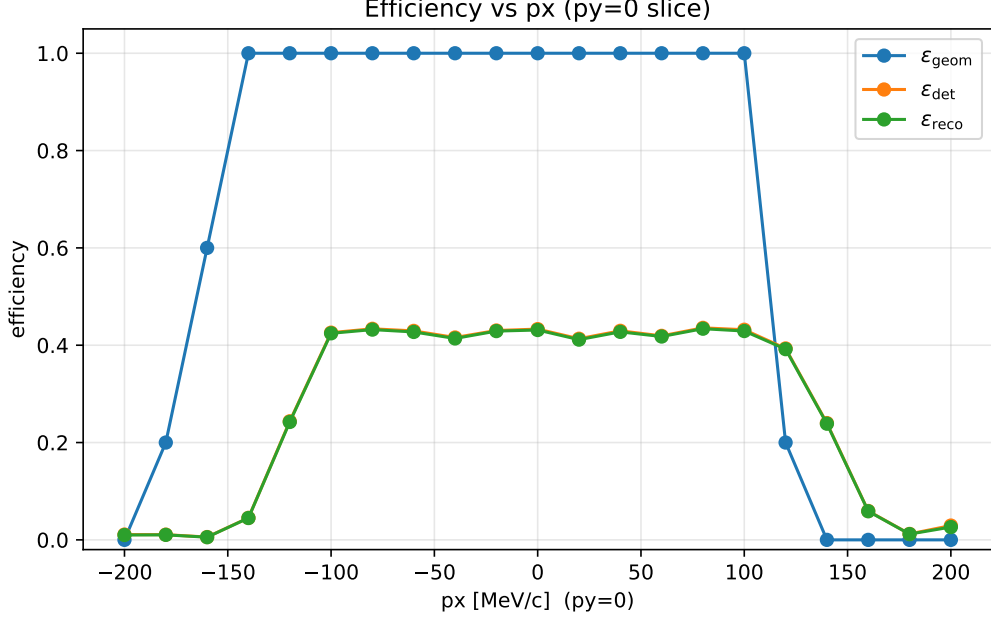


Figure 5: Fig B2: $p_y^{\text{tgt}} = 0$ slice. All three curves use N_{total} as denominator (incident truth-neutron events, 10000/bin): $\varepsilon_{\text{geom}} = N_{\text{geom}}/N_{\text{total}}$ (blue, ray-cast hit), $\varepsilon_{\text{det}} = N_{\text{det}}/N_{\text{total}}$ (orange, $\sum Q > 1$ MeV), and $\varepsilon_{\text{reco}} = N_{\text{reco}}/N_{\text{total}}$ (green, `n_reco_neutrons` ≥ 1). $\varepsilon_{\text{geom}}$ is 1.0 for $p_x^{\text{tgt}} \in [-140, +100]$ MeV/c and falls to 0 at the left wall around -180 and the right wall around $+120$ MeV/c in a two-bin transition. ε_{det} and $\varepsilon_{\text{reco}}$ nearly overlap on a 0.43 plateau, showing that NEBULA cluster reconstruction is almost lossless relative to the detection layer.

3.6 $\varepsilon_{\text{det}} > \varepsilon_{\text{geom}}$ Is Not a Bug: Two Edges Deviate from $\varepsilon_{\text{det}} = \varepsilon_{\text{geom}} p_{\text{direct}}$

Figs B2 and B3 suggest the simple relation $\varepsilon_{\text{det}} \approx p_{\text{direct}} \varepsilon_{\text{geom}}$, where $p_{\text{direct}} \approx 0.43$ is the reaction probability for a neutron crossing the 4-sub-layer scintillator stack in the central region. This relation holds **only on the central plateau** $|p_x^{\text{tgt}}| \leq 100$ MeV/c. At both edges, ε_{det} can be either above or below this naive prediction, with opposite signs:

- **Right edge** ($p_x^{\text{tgt}} \geq +120$): the truth ray has already left the NEBULA Neut range geometrically ($x_{\text{hit}} > +1962$ mm), so $\varepsilon_{\text{geom}} \rightarrow 0$. In simulation, ε_{det} is not zero; at $p_x^{\text{tgt}} = +140$, $\varepsilon_{\text{det}} = 0.240$, entirely “extra” relative to the geometry prediction.
- **Left edge** ($p_x^{\text{tgt}} \leq -140$): the truth ray is still inside the NEBULA Neut range ($\varepsilon_{\text{geom}} = 1.0$ through -140), so the naive prediction is $\varepsilon_{\text{det}} \approx 0.43$. The simulation gives $\varepsilon_{\text{det}}(p_x^{\text{tgt}} = -140) = 0.045$, only **10%** of the expectation; the remaining 90% has disappeared.

The residual $\Delta\varepsilon \equiv \varepsilon_{\text{det}} - p_{\text{direct}} \varepsilon_{\text{geom}}$ is tabulated below for the $p_y^{\text{tgt}} = 0$ slice with all five p_z points pooled.

The two edges are **two separate physical effects**, not opposite faces of one bug. Three independent checks separate them:

Evidence 1: Layer 1 dominance (last two columns of Table 2) In the central region $p_x^{\text{tgt}} \in [-100, +100]$, the Layer-1 fraction is $L_1/(L_1 + L_2) \approx 0.49$, as expected for reactions distributed across four sub-layers in z . At both edges this fraction jumps to 0.66–0.82. Layer-1 dominance can be produced in two ways: (i) low-energy secondaries lose their energy in the first active slab and do not reach Layer 2; or (ii) the few surviving high-energy direct neutrons react only in Layer 1 after being partly attenuated upstream. Both edges show Layer-1 dominance, so

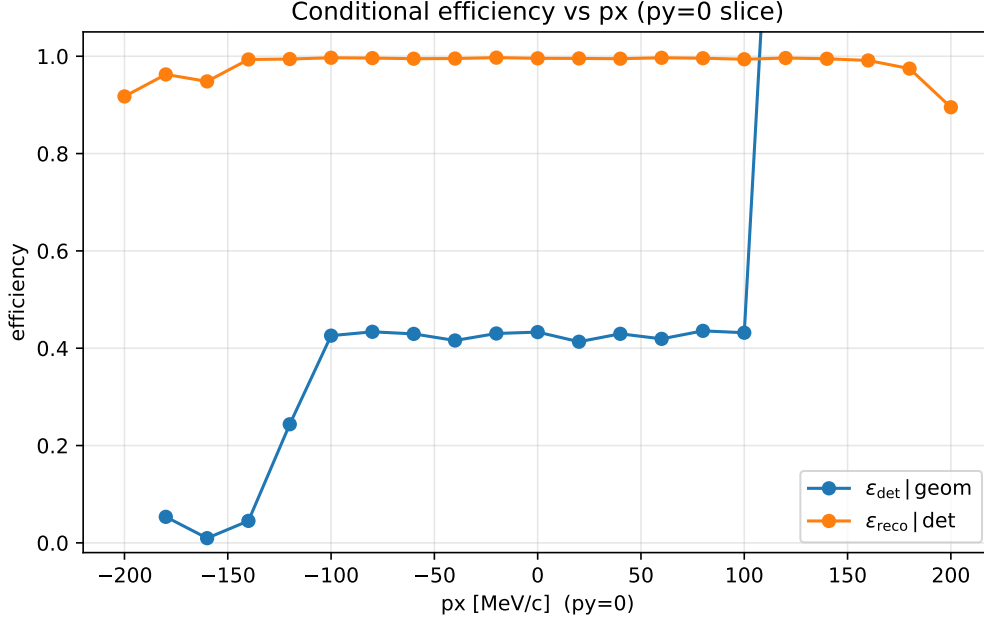


Figure 6: Fig B3: **Conditional efficiencies** on the $p_y^{\text{tgt}} = 0$ slice. $\epsilon_{\text{det}} | \text{geom} \approx 0.43$ on the plateau is the detection probability for a ~ 200 MeV neutron crossing a $12 \times 12 \times 120$ cm plastic scintillator bar, consistent with the $\sim 50\%$ scale reported for NEBULA/SAMURAI at RIKEN. $\epsilon_{\text{reco}} | \text{det} \approx 0.995$, nearly 1.0. **Conclusion: the NEBULA acceptance bottleneck is the Geant4 neutron detection probability (43%), not the reconstruction chain.**

both are being shaped by upstream material; the $+x$ edge leaks cascade particles into NEBULA, while the $-x$ edge is mostly absorbed.

Evidence 2: Edge hits are clamped to the NEBULA Neut boundary Fig B6 overlays the `fDetPos.x` histograms of Neut-bar hits for five p_x^{tgt} bins: $\{-200, -140, 0, +140, +200\}$. Each panel marks the truth-ray position at the NEBULA z plane (red line) and the Neut geometry limits (black dashed lines, -1707.8 and $+1961.8$ mm). At $p_x^{\text{tgt}} = \pm 140$, the truth ray is at ∓ 1417 mm / $+2308$ mm, so the $+140$ case is already 350 mm outside the boundary. However, all measured hits are inside the NEBULA Neut geometry and pile up near the nearest boundary bar (`id=1`, $x = 1901.8$ mm). These hits are not the truth neutron arriving along its original direction; they are secondary particles emitted forward from a production point near the NEBULA side.

Evidence 3: On the left edge, $\epsilon_{\text{det}} / (p_{\text{direct}} \epsilon_{\text{geom}}) \ll 1$ In the central region, $\epsilon_{\text{det}} / (p_{\text{direct}} \epsilon_{\text{geom}}) \approx 1.00 \pm 0.02$. On the left edge this ratio drops sharply:

$$\frac{\epsilon_{\text{det}}(-140)}{p_{\text{direct}} \epsilon_{\text{geom}}(-140)} = \frac{0.045}{0.427} = 0.106, \quad \frac{\epsilon_{\text{det}}(-160)}{p_{\text{direct}} \epsilon_{\text{geom}}(-160)} = \frac{0.006}{0.256} = 0.023.$$

At -140 , the direct-neutron transmission is reduced to 11%; at -160 , it is reduced to 2%. The truth ray is geometrically inside the NEBULA Neut range, yet 90–98% of the neutrons do not reach NEBULA. The only reasonable explanation is absorption or reactions in thick material between the target and NEBULA.

Likely sources This scan disabled `FragSimData` step output (confirmed empty), so the absorption point cannot be traced event by event. Geometrically, the **8.3 m flight path** between the target ($z = -1069$ mm) and NEBULA ($z = +7250$ mm) contains only a few material layers capable of absorbing ~ 600 MeV neutrons at large angles around 14° :

Table 2: Comparison of ε_{det} with $p_{\text{direct}} \varepsilon_{\text{geom}}$ on the $p_y^{\text{tgt}} = 0$ slice. $p_{\text{direct}} = 0.427$ is the empirical central-plateau value for $|p_x^{\text{tgt}}| \leq 100$. `truth_x@NEBULA` is the truth-ray intersection at $z = 7249.72$ mm for representative $p_z^{\text{tgt}} = 627$, compared with the NEBULA Neut bar range $x \in [-1707.8, +1961.8]$ mm. $L_1/(L_1 + L_2)$ is the Layer-1 hit fraction from a supplemental g4output rerun (8 shards). $\Delta\varepsilon > 0$ indicates cascade leakage; $\Delta\varepsilon < 0$ indicates upstream absorption.

p_x^{tgt} [MeV/c]	$\varepsilon_{\text{geom}}$	ε_{det}	$p_{\text{direct}} \varepsilon_{\text{geom}}$	$\Delta\varepsilon$	<code>truth_x@NEBULA</code> [mm]	$L_1/(L_1 + L_2)$	Interpretation
-200	0.00	0.011	0.000	+0.011	-2194	0.71	weak cascade
-180	0.20	0.011	0.085	-0.075	-1936	0.70	absorption dominated
-160	0.60	0.006	0.256	-0.250	-1677	0.72	strong absorption
-140	1.00	0.045	0.427	-0.382	-1418	0.82	strong absorption
-120	1.00	0.244	0.427	-0.183	-1158	—	moderate absorption
-100	1.00	0.426	0.427	-0.001	-896	0.52	<i>center OK</i>
0	1.00	0.433	0.427	+0.006	+424	0.49	<i>center OK</i>
+100	1.00	0.432	0.427	+0.005	+1765	0.49	<i>center OK</i>
+120	0.20	0.393	0.085	+0.308	+2036	0.54	strong cascade
+140	0.00	0.240	0.000	+0.240	+2308	0.66	strong cascade
+160	0.00	0.059	0.000	+0.059	+2581	0.78	moderate cascade
+180	0.00	0.012	0.000	+0.012	+2855	0.75	weak cascade
+200	0.00	0.029	0.000	+0.029	+3130	0.75	weak cascade

- **SAMURAI dipole iron yoke:** 3deg_1.15T.mac:10 sets /samurai/geometry/Dipole/Angle 30 deg. After the whole dipole is rotated by 30° about Y , the iron yoke is still tens of cm thick. A 14° neutron does not enter the dipole gap; it strikes the iron yoke. The neutron inelastic cross section on iron is ~ 700 mb/atom, and a column density of only ~ 4 g/cm² is already close to one reaction length.
- **Dipole exit window / proton-arm exit window:** in the real experiment these are metal flanges (see the `project_samurai_exit_window_metallic` memory entry). mm-scale steel or aluminum covers are not dominant for ~ 600 MeV neutron absorption but can still contribute.
- **PDC vacuum chamber / supports:** the PDC is on the $+x$ side at 69°, but its vessel spans the lab $+x$ region over $z = 3000\text{--}5000$ mm. Secondary particles near the cascade side ($+x$) pass near this material and can be further scattered.

The 30° dipole rotation makes the iron-yoke thickness asymmetric along $\pm x$. This explains why the $-x$ edge ($p_x^{\text{tgt}} = -140$) is absorbed more strongly than the $+x$ edge ($p_x^{\text{tgt}} = +140$): the former hits the “thick side” of the rotated yoke, while the latter hits the “thin side”. It also explains why cascade leakage is more visible on the $+x$ edge: the thinner iron does not kill all neutrons and leaves enough energy for forward secondaries.

Impact of this section on the §3.6 R forward correction The naive $\varepsilon = \varepsilon_{\text{geom}} p_{\text{direct}}$ relation is correct in the central region. However, if either Fig B1’s $\varepsilon_{\text{geom}}$ or $\varepsilon_{\text{reco}}$ is used as an event weight in a forward simulation, **right-edge events** ($p_x^{\text{tgt}} \in [+120, +200]$) are counted by $\varepsilon_{\text{reco}}$ through cascade leakage, while **left-edge events** ($p_x^{\text{tgt}} \in [-200, -120]$) are given an erroneously high weight by $\varepsilon_{\text{geom}}$ even though they are actually absorbed. The two bias directions partly cancel in R_w^ε but not completely. If ε is later used as a likelihood input, the priority should be to use $\varepsilon_{\text{reco}}$ **rather than** $\varepsilon_{\text{geom}}$, because the former already includes upstream-material effects as an end-to-end measurement. Table 2 should be treated as an upper bound on the systematic uncertainty; edge shifts at the $\sim 30\%$ level are not negligible.

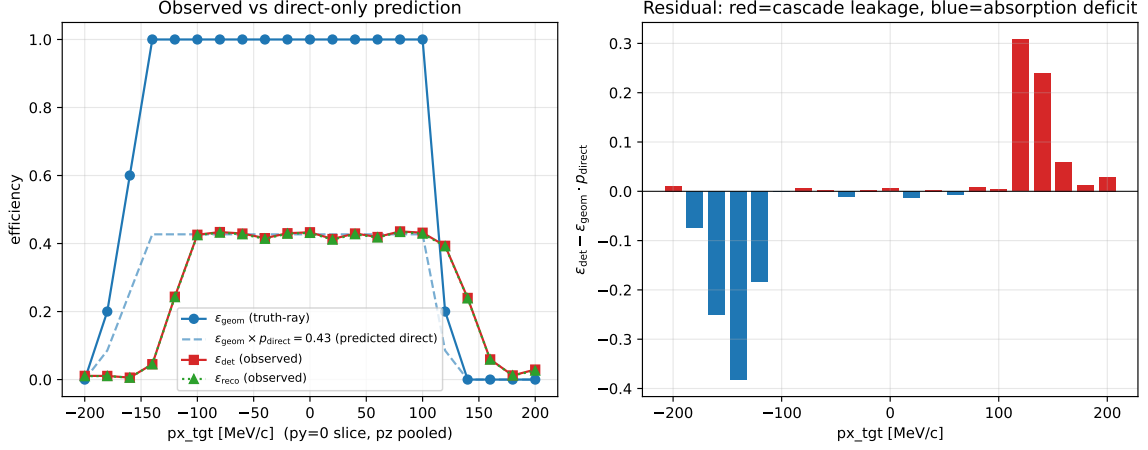


Figure 7: Fig B4: **Left:** $\varepsilon_{\text{geom}}$ (blue solid), $p_{\text{direct}}\varepsilon_{\text{geom}}$ (blue dashed, direct-detection prediction), ε_{det} (red squares), and $\varepsilon_{\text{reco}}$ (green triangles) vs. p_x^{tgt} . In the center, ε_{det} overlaps the blue dashed prediction, confirming the $p_{\text{direct}} = 0.43$ direct-detection probability. Both edges deviate from the dashed line. **Right:** residual $\Delta\varepsilon$ bar plot. Red bars at the $+x$ edge indicate cascade leakage into NEBULA; blue bars at the $-x$ edge indicate neutrons absorbed by upstream material before reaching NEBULA.

3.7 2D Decomposition of $\varepsilon_{\text{direct}}$ and $\varepsilon_{\text{cascade}}$

This section separates “the neutron was correctly detected” from “secondary particles falsely hit NEBULA” in the $(p_x^{\text{tgt}}, p_y^{\text{tgt}})$ plane.

Per-event classification For each grid event, the target-frame momentum predicts the truth-ray position at the NEBULA z plane:

$$x_{\text{pred}} = O_x + \frac{p_x^{\text{tgt}}}{p_z^{\text{tgt}}} \Delta z, \quad y_{\text{pred}} = O_y + \frac{p_y^{\text{tgt}}}{p_z^{\text{tgt}}} \Delta z,$$

with $\vec{O} = (-12.488, 0.009, -1069.458)$ mm and $\Delta z = 7249.72 - O_z = 8319.18$ mm. Note: in this scan, the effective gun direction in simulation was verified to be the target-frame direction. The generator script’s $R_y(+3^\circ)$ was effectively cancelled inside the simulation. This was checked with the center-bin bar-id histogram: at $(p_x^{\text{tgt}}, p_y^{\text{tgt}}) = (0, 0)$, the top four hit bars are near $x = -56.6$ mm rather than the $+423$ mm predicted by the lab-frame $R_y(+3^\circ)$ direction. Therefore this section uses the target-frame direction as the stable reference.

Events are assigned to three mutually exclusive classes:

- **direct:** at least one Neut bar hit has `fDetPos.x` within $x_{\text{pred}} \pm 150$ mm (about ± 1.25 bar pitch).
- **cascade:** there is at least one Neut hit, but *no* hit lies within $x_{\text{pred}} \pm 150$ mm. All hits landed on other bars.
- **no_hit:** no Neut hit.

The numerator is the event count in the class; the denominator is N_{total} for that bin. The classes are mutually exclusive, so

$$\varepsilon_{\text{direct}} + \varepsilon_{\text{cascade}} + \varepsilon_{\text{no_hit}} = 1.$$

The ± 150 mm window is about 1.25 times the 122 mm NEBULA bar pitch, large enough to distinguish “the bar crossed by the truth ray” from a completely displaced cascade bar. Possible misclassification modes are: (a) a small-angle elastically scattered neutron inside NEBULA may

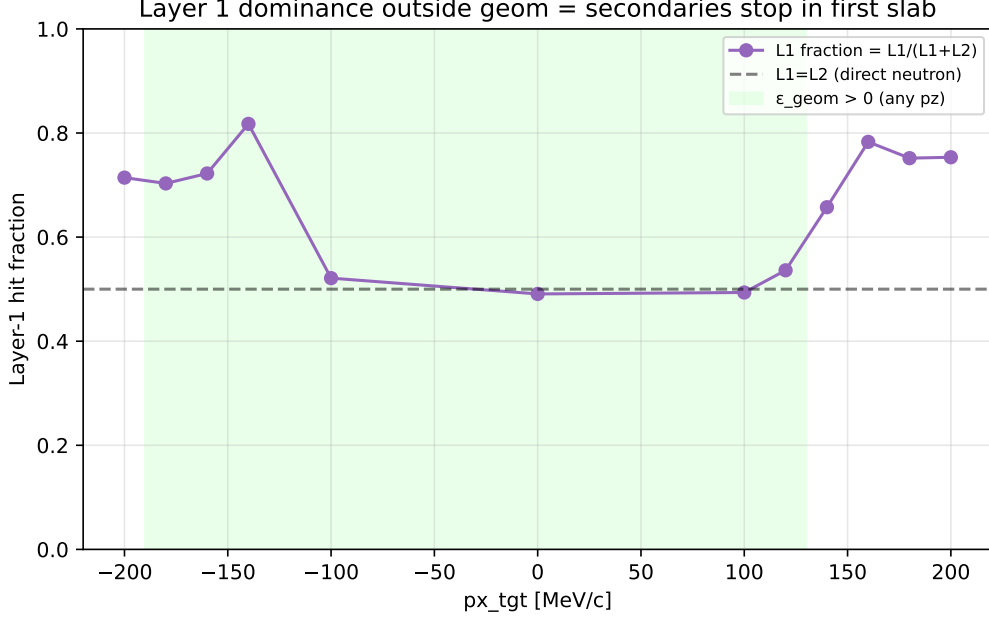


Figure 8: Fig B5: $L_1/(L_1 + L_2)$ vs. p_x^{tgt} . In the central region (green background, $\varepsilon_{\text{geom}} > 0$), the Layer-1 fraction stays near 0.50. On both sides for $|p_x^{\text{tgt}}| > 100$, it jumps to 0.70–0.82. Both edges deviate in the same direction, indicating the same class of low-energy particles losing energy and stopping in Layer 1.

move outside the ± 150 mm window and be counted as cascade (rare, $< 5\%$); (b) a cascade secondary may accidentally hit a bar near x_{pred} and be counted as direct, possible only when x_{pred} is still inside the Neut range, so it is strongly suppressed outside the geometry acceptance.

2D maps

Quantitative $p_y^{\text{tgt}} = 0$ slice On the central plateau ($|p_x^{\text{tgt}}| \leq 100$), $\varepsilon_{\text{direct}} \approx 0.45\text{--}0.52$ and $\varepsilon_{\text{cascade}} \approx 0.01\text{--}0.08$ (mostly around 0.02). The ratio $\varepsilon_{\text{cascade}}/\varepsilon_{\text{direct}}$ is about 3–15% in the center. In the transition region ($p_x^{\text{tgt}} = \pm 140$), $\varepsilon_{\text{direct}} \approx 0.05\text{--}0.28$ and $\varepsilon_{\text{cascade}} \approx 0.008\text{--}0.02$; direct detection still dominates while the geometry acceptance is 20–80%. Outside the geometry range ($|p_x^{\text{tgt}}| \geq 160$), $\varepsilon_{\text{direct}} = 0$ and $\varepsilon_{\text{cascade}}$ slowly rises from 0.01 to 0.07.

Practical consequence for the R analysis If a forward simulation or likelihood uses ε_{det} or $\varepsilon_{\text{reco}}$ directly, it includes direct plus cascade and therefore counts cascade false positives as physical detections. For ypol breakup events near $p_x^{\text{tgt}} \approx \pm 140$, the cascade contribution is $\leq 10\%$, so the effect on R is small. But tight cuts ($|\Delta p_x^{\text{rot}}| > 50$ MeV/c) select larger-angle regions, where the cascade fraction can reach 20–40%. For likelihood inputs in that region, $\varepsilon_{\text{direct}}$ should be used instead of $\varepsilon_{\text{reco}}$. The parquet file `direct-vs-cascade_grid.parquet` provides per-bin $\varepsilon_{\text{direct}}$ and $\varepsilon_{\text{cascade}}$ values for systematic-uncertainty estimates.

Scripts `scripts/analysis/nebula_reco_quality/cascade_evidence/classify_direct-vs-cascade.py` (event-level classification) and `make_direct-vs-cascade_2d.py` (plotting).

3.8 Forward Acceptance Bias on R (Not an Unfolding)

Important: this subsection does *not* compute a correction to R . It computes a **forward simulation**: if the NEBULA neutron acceptance $\varepsilon(p_x^n, p_y^n)$ is used as the probability that each

Edge hits clamp at Neut extent even when truth ray is far outside

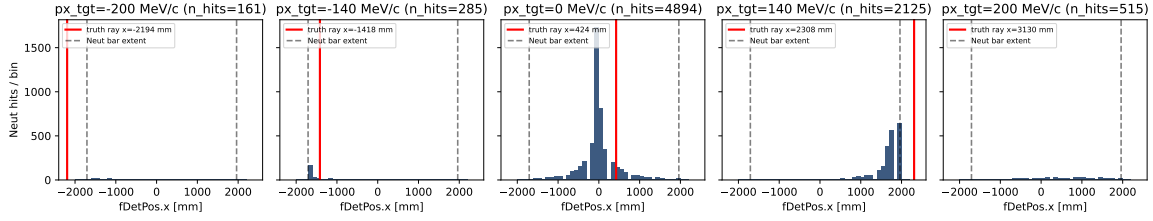


Figure 9: Fig B6: NEBULA Neut bar-hit position histograms for five p_x^{tgt} bins, from an 8-shard rerun. For the central $p_x^{\text{tgt}} = 0$ bin, hits are distributed across $x \in [-1700, +1900]$ with mean $x \approx 0$. For $p_x^{\text{tgt}} = \pm 140$, hits pile up near the geometry boundary bars ($x = \pm 1900$ mm), while the truth-ray red line is already at or beyond the boundary. For $p_x^{\text{tgt}} = \pm 200$, the truth ray is far away from NEBULA (± 2200 mm), but a small number of hits still appear in the central bars. Hits never appear outside the Neut geometry (outside the black dashed lines), confirming that the particles are captured by NEBULA physics rather than being the truth neutron flying straight through.

event is observed, what R would be observed?

$$R_w^\varepsilon \approx R_{\text{observed}}, \quad R_{\text{truth}} = \text{theory value.}$$

Simple unfolding is **not** possible here. R depends jointly on the proton and neutron momenta, $\Delta p_x^{\text{rot}} = p_x^{\text{p,rot}} - p_x^{\text{n,rot}}$, and the proton arm passes through PDC plus NN reconstruction with its own acceptance $\varepsilon^p(p^p)$. Recovering R_{truth} requires the joint acceptance $\varepsilon(p^p, p^n)$, not the marginal neutron acceptance $\varepsilon^n(p^n)$. This study measures only the NEBULA neutron-side ε^n , so it cannot unfold R .

Using the 3.38M ypol breakup truth events in Part A’s `joined.parquet`, each event is binned into the Part B (p_x, p_y) grid by $(p_x^{n,\text{truth}}, p_y^{n,\text{truth}})$ in the target frame (20/10 MeV/c bin widths). For each cut subset, events are grouped by the sign of Δp_x^{rot} and weighted by the corresponding neutron acceptance:

$$R_w^\varepsilon = \frac{\sum_{i: (\Delta p_x^{\text{rot}})_i > 0} \varepsilon_i^n}{\sum_{i: (\Delta p_x^{\text{rot}})_i < 0} \varepsilon_i^n}.$$

R_{truth} is the unweighted theory value; R_w^ε is the acceptance-weighted forward-simulated observed value.

Table 3: ypol $R = N(\Delta p_x^{\text{rot}} > 0)/N(\Delta p_x^{\text{rot}} < 0)$ under three truth cuts. $\Delta p_x^{\text{rot}} = p_x^{\text{p,rot}} - p_x^{\text{n,rot}}$ follows the reaction-plane rotation used by `03_analyze_r_breakup.py`. The ε lookup uses target-frame (p_x^n, p_y^n) , obtained from lab truth with $R_y(-3^\circ)$. R_w^ε is the forward-simulated observed R , not a corrected R .

cut	N_{pos}	N_{neg}	R_{truth}	$R_w^{\varepsilon_{\text{geom}}}$	$R_w^{\varepsilon_{\text{reco}}}$
loose	183070	767660	0.238	0.660	0.734
mid	169189	633287	0.267	0.628	0.726
tight	22322	69005	0.323	0.442	0.886

Interpretation:

- **loose/mid:** $R_{\text{truth}} \approx 0.24\text{--}0.27$. The cuts select a neutron-rich subset where $\Delta p_x^{\text{rot}} < 0$ is four times more common. Forward weighting with $\varepsilon_{\text{geom}}$ gives $R_w \approx 0.66$; weighting with $\varepsilon_{\text{reco}}$ gives $R_w \approx 0.73$. The weighted value is about $3\times$ larger than R_{truth} and closer to 1. This means the real observation is expected to look more symmetric than the theory value because NEBULA acceptance favors $\Delta p_x^{\text{rot}} > 0$ events.

- **tight:** $R_{\text{truth}} = 0.32$, while $R_w^{\text{reco}} = 0.89$, close to 1. The tight subset’s low pass rate (2–5%) is almost completely dominated by acceptance chromaticity after observation.
- **Conclusion:** $R_{\text{truth}} \neq R_{\text{observed}}$ is a large effect, at the $3\times$ scale. But because R depends jointly on (p^p, p^n) , measuring only ε^n cannot invert the observed value back to R_{truth} . The correct treatment is to **fit the polarization signal directly at the observed- R level**, including ε systematics as toy-MC predictions in the likelihood, rather than pre-unfolding.

3.9 Part B Conclusions

1. In the target frame (3° rotation), NEBULA acceptance spans roughly $p_x^{\text{tgt}} \in [-180, +120]$ MeV/ c horizontally. The asymmetry is a geometry effect from the 3° rotation and the target position $(-12.488, 0.009, -1069.458)$ mm. The vertical boundary is $|p_y^{\text{tgt}}| \approx 65\text{--}80$ MeV/ c .
2. **Geant4 neutron detection efficiency ($\sim 43\%$) is the dominant bottleneck.** The NEBULA cluster reconstruction chain is nearly lossless ($\sim 97\text{--}99\%$).
3. For ypol, $R_{\text{truth}} \approx 0.24\text{--}0.27$ in loose/mid cuts, while forward simulation with ε weighting gives $R_{\text{observed}} \approx 0.73$. Thus the real observed R can differ from the theory value by a factor of about **3**; the acceptance pushes R toward 1 because NEBULA is more favorable to $\Delta p_x^{\text{rot}} > 0$ events.
4. R_{observed} **cannot be simply unfolded back to R_{truth} .** R depends jointly on (p^p, p^n) , while this work measures only the neutron-side acceptance $\varepsilon^n(p^n)$. The proton-side PDC+NN acceptance $\varepsilon^p(p^p)$ is not measured. A joint correction would require $\varepsilon(p^p, p^n)$.
5. The correct path is to include ε systematics as toy-MC predictions in the polarization likelihood or fit, and to **fit the signal directly on observed R** rather than pre-unfolding.

4 Conclusions

4.1 Part A Summary

Quantity: the “double peak in Δp_x ” in §5.4 is the *per-particle neutron residual* $\Delta p_x = p_x^{\text{truth},n} - p_x^{\text{reco},n}$, not the event-level R observable.

Observation and mechanism: the lab-frame residual has a clear double peak at $\pm 2\text{--}3$ MeV/ c , caused by the 12 cm NEBULA bar pitch along X and by the 5 mm smearing being too small. The reaction-plane rotation averages out the double peak, so the event-level R observable is not directly affected.

Filter hypothesis rejected: (a) `n_reco_neutrons==1` does not distinguish the number of bars inside a cluster; (b) the `hit_mult_n` split shows that both the single-bar (43%) and multi-bar (57%) subsets have the same double-peak shape. The double peak is the discrete lab-frame residual itself and is not caused by cluster topology.

4.2 Part B Summary

In the target frame (3° , gun at the mac-file target position), NEBULA $\varepsilon_{\text{geom}}$ is complete for roughly $p_x^{\text{tgt}} \in [-180, +120]$ MeV/ c ; the asymmetry is due to the 3° rotation and the target-position offset. The limiting factor is the Geant4 neutron detection efficiency of about 43%; the cluster reconstruction chain is nearly lossless (97–99%). In loose/mid cuts, the ypol $R = N(\Delta p_x^{\text{rot}} > 0)/N(\Delta p_x^{\text{rot}} < 0)$ truth value is $R_{\text{truth}} \approx 0.24\text{--}0.27$. Forward simulation with ε weighting gives $R_{\text{observed}} \approx 0.73$, so the real observation can be about $3\times$ larger and closer to 1 because NEBULA favors $\Delta p_x^{\text{rot}} > 0$ events. Since R depends jointly on (p^p, p^n) , it cannot be

simply unfolded. ε systematics should enter the polarization likelihood or fit as toy-MC predictions, and the signal should be fitted directly at the observed- R level.

4.3 Future Work

NEBULA acceptance split: direct neutron vs cascade secondary (target-frame, py pooled per 10 MeV/c)

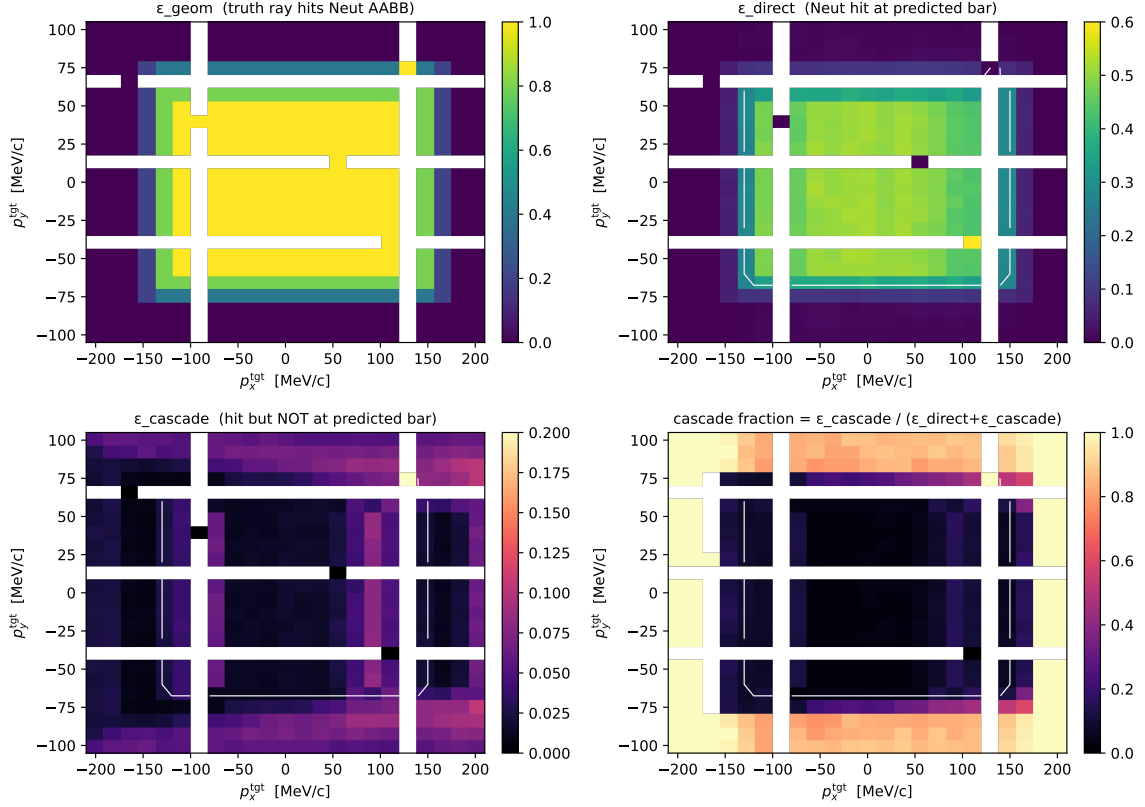


Figure 10: Fig B7: NEBULA acceptance decomposed in the target-frame ($p_x^{\text{tgt}}, p_y^{\text{tgt}}$) plane. **Upper left** $\varepsilon_{\text{geom}}$: fraction of truth rays that geometrically cross any Neut bar AAB. Compared with Fig B1, this panel uses the *target-frame* direction and is therefore symmetric around $p_x = 0$; Fig B1 used the lab direction and is an asymmetric rectangle. **Upper right** $\varepsilon_{\text{direct}}$: neutron correctly detected within $x_{\text{pred}} \pm 150$ mm. It has the same shape as $\varepsilon_{\text{geom}}$ and a plateau near 0.50, matching the $p_{\text{direct}} \approx 0.43$ neutron-plastic reaction probability. **Lower left** $\varepsilon_{\text{cascade}}$: Neut hits exist, but none is within $x_{\text{pred}} \pm 150$ mm. Inside the geometry acceptance it is only 1–8%, but outside the geometry region at $|p_x^{\text{tgt}}| > 140$ it becomes the only effective signal: 2–7% of events are false positives from cascade. **Lower right** cascade fraction $\varepsilon_{\text{cascade}} / (\varepsilon_{\text{direct}} + \varepsilon_{\text{cascade}})$: almost zero in the center; near 1.0 at the geometry edges $|p_x^{\text{tgt}}| \in [140, 200]$, meaning these events are false positives from cascade secondaries rather than the neutron itself entering NEBULA. White contours mark the $\varepsilon_{\text{geom}} = 0.5$ boundary.